

Attention Mechanisms in Computer Vision: A Comprehensive Analysis

Authors: Dr. James Liu, Prof. Anna Kowalski, Dr. Robert Kim

Conference: CVPR 2025

ABSTRACT:

We present a thorough analysis of attention mechanisms in computer vision, comparing Vision Transformers (ViTs), Convolutional Attention, and hybrid architectures across 12 benchmark datasets.

1. BACKGROUND

Attention mechanisms, originally developed for NLP, have been successfully adapted for computer vision tasks. The Vision Transformer (ViT) treats images as sequences of patches, applying self-attention globally.

2. EXPERIMENTAL SETUP

Datasets: ImageNet-1K, CIFAR-100, MS-COCO, ADE20K

Models tested:

- ViT-Base, ViT-Large, ViT-Huge
- ResNet + CBAM (Convolutional Block Attention Module)
- Swin Transformer variants
- ConvNeXt with attention layers

Training details:

- Batch size: 256
- Learning rate: $1e-4$ with cosine annealing
- Data augmentation: RandAugment, MixUp, CutMix
- Hardware: 8x A100 GPUs

3. RESULTS

Top-1 Accuracy on ImageNet-1K:

- ViT-Large: 85.2%
- Swin-Large: 86.4%
- ConvNeXt-Large: 87.1%

Key insights:

- Hierarchical attention (Swin) outperforms global attention (ViT)
- Hybrid CNN-Transformer models achieve best efficiency
- Attention visualization reveals focus on object boundaries

4. CONCLUSION

The future of computer vision lies in carefully designed attention mechanisms that balance global and local information.